Machine Learning Methods for n-Colorant Printer Characterization: A Corrected and Extended Evaluation

Muhammad Hamza Zafar[1,*] Philip J. Green[1]

[1] Norwegian Colour and Visual Computing Laboratory, Department of Computer Science, Norwegian University of Science and Technology (NTNU), Gjøvik, Norway
[*] Correspondence: muhammad.h.zafar@ntnu.no

Abstract

[TODO: Abstract stub – fill in once Results/Discussion are drafted (Task 2). MDPI Technologies caps the abstract at 200 words; single paragraph, no citations, no undefined abbreviations.] Printer color characterization maps a device's colorant inputs (CMY, CMYK, or higher-channel combinations) to device-independent color coordinates (CIE XYZ/Lab). A prior conference study (AIC 2025) compared 14 machine learning methods against third-order polynomial regression for $n = 3$ (CMY) inputs across three printer datasets, finding no consistent advantage for the learned models. That analysis is superseded here: its CIEDE2000 errors were computed on normalized rather than physical XYZ values, understating errors by roughly a factor of five. This paper reports corrected $n = 3$ and $n = 4$ (CMYK) results across three printer datasets; evaluates cross-printer generalization on 13 white-backing newsprint press runs under within-run, cross-run, and leave-one-run-out training conditions; tests whether refining the classical third-order polynomial's coefficients directly on ΔE_{00} (instead of least-squares on XYZ) improves worst-case accuracy; and benchmarks a frontier large language model (GPT-4o) as a direct color predictor against the full classical/ML model set. **[RESULTS-PENDING: n>4 (CMYKOGV) results, if the dataset lands in time]**. **[TODO: One-sentence headline finding once available.]**

Keywords: printer characterization; color management; CIEDE2000; machine learning; Gaussian process regression; polynomial regression; CMYK; n-colorant printing; ICC profiling

1 Introduction

[TODO: Extend the AIC 2025 introduction (Task 2, per docs/plans/07-paper.md). Planned additions: (i) the lookup-table-inefficiency motivation for higher-channel printing (Phil, Apr 2026 meeting) – LUTs scale combinatorially with the number of colorants, so device characterization methods that avoid dense LUT construction become increasingly attractive as n grows beyond 4; (ii) restate Phil Green's two core questions verbatim: (a) can ML/AI match or beat existing methods for $n \leq 4$ colorants, avoiding the need for gray-component-replacement-style algorithms; (b) can AI handle $n > 4$ colorant systems (e.g. CMYKOGV) where classical polynomial methods are not designed to scale; (iii) state explicitly that this paper corrects and supersedes the AIC 2025 conference numbers (forward-reference to the Methods correction paragraph).]

2 Related Work

[TODO: Task 2. Prior device-characterization literature already in refs.bib (Bala 2003; Cheung et al. 2004; Kang & Anderson 1992; Tominaga 1993) plus the per-method ML references (Breiman 1984/2001; Cortes & Vapnik 1995; Cover & Hart

1967; Friedman 2001; Geladi & Kowalski 1986; Hoerl & Kennard 1970; Rasmussen & Williams 2006; Tibshirani 1996; Zou & Hastie 2005). Add: Kiran's Optics Express n-colour paper and the PhD-work benchmark Phil pointed to (both currently commented placeholders in refs.bib – exact citations pending); ISO/TC 130 standardization context (Pang 2024); the FOGRA51 characterization dataset (Fogra 2020); the colour-science library used throughout this pipeline (placeholder in refs.bib, pending exact version/DOI).]

3 Methods

3.1 Datasets

Three printer characterization datasets are used, each measured at 1,617 CMYK combinations under a shared measurement protocol (spectrophotometric CIE Lab and CIE XYZ, D50 illuminant, 2° standard observer). Each dataset is evaluated twice: once restricted to the $K = 0$ subset with only C, M, Y as inputs (818 rows – the $n = 3$ / CMY condition), and once using the full sample set with C, M, Y, K as four inputs (1,617 rows – the $n = 4$ / CMYK condition). Unlike the AIC 2025 conference version, the K channel is never dropped while its rows are kept: doing so maps identical CMY inputs to different XYZ outputs whenever K differs, which is an internal label inconsistency, not a modeling choice.

Table 1: Datasets used in the $n = 3$ and $n = 4$ conditions. Row counts are pooled across all 5 cross-validation folds (Section 3.3), so every sample is predicted exactly once.

Dataset	Substrate	Inputs (n)	Rows	Source
PC10	Cardboard	CMY	818	APTEC PC10 (2023)
PC10	Cardboard	CMYK	1,617	APTEC PC10 (2023)
PC11	Coated paper (CCNB)	CMY	818	APTEC PC11 (2023)
PC11	Coated paper (CCNB)	CMYK	1,617	APTEC PC11 (2023)
FOGRA51	Reference (ISO 12647-2)	CMY	818	Fogra Research Institute [1]
FOGRA51	Reference (ISO 12647-2)	CMYK	1,617	Fogra Research Institute [1]

A fourth dataset is used for the cross-printer generalization experiments of Section 4.4: IFRA 2005 newsprint, comprising 13 white-backing (wb) press runs, each with 1,485 CMYK samples measured as 36-band spectral reflectance (380–730 nm) and converted to CIE XYZ under the D50 illuminant and 2° standard observer via the `colour-science` library. The complementary 30 black-backing (bb) runs are not used in this paper: a chart-layout mismatch between the wb and bb measurement files is still being resolved with the data provider. Per Phil's instruction, wb and bb are kept as separate conditions rather than pooled, so this exclusion is scoped to bb only and does not affect the wb results reported here.

[TODO: Task 06: append a CMYKOGV ($n = 7$) row to Table 1 if that dataset arrives from Phil/Will in time. Append the 30 IFRA black-backing runs once the chart-layout mismatch is resolved.]

3.2 Models

Fourteen regression methods are compared, matching the AIC 2025 study's model set with two configuration corrections noted below. All models are implemented input-dimension-agnostically, so the same code path serves the $n = 3$, $n = 4$, and (if it lands) $n = 7$ conditions.

[TODO: Task 2: add explicit equations for each model, in particular the corrected 3rd-order polynomial expansion (Phil flagged an "odd expression" in the AIC ver-

Table 2: Model registry (14 methods). Configurations match the AIC 2025 paper's best configs except where marked (*), which were degenerate there (Lasso/Elastic Net at $\alpha = 1.0$ collapsed to a constant predictor; SVR at $\varepsilon = 0.1$ produced an insensitivity tube covering 10% of the target range) and are replaced with standard, non-degenerate values.

Model	Configuration
Polynomial regression (degree 3)	Full 3rd-order feature expansion + ordinary least squares
Ridge regression	$\alpha = 0.5$
Lasso regression*	$\alpha = 10^{-3}$
Elastic Net*	$\alpha = 10^{-3}$, ℓ_1 ratio $= 0.5$
Principal component regression	All components retained + Ridge ($\alpha = 0.5$)
Partial least squares regression	3 components
k-nearest neighbors	$k = 5$, uniform weights
Support vector regression*	RBF kernel, $C = 10$, $\gamma =$ scale, $\varepsilon = 0.01$
Decision tree	Unconstrained depth
Random forest	200 trees, max depth 15
Gradient boosting	200 estimators, learning rate 0.05, max depth 5
Gaussian process regression	Const $\times$ RBF + WhiteKernel(10^{-5}), y-normalized
MLP (shallow)	1 hidden layer, 64 units, L-BFGS
MLP (deep)	3 hidden layers, 64 units each, L-BFGS

sion's equation to fix). Every stochastic model above is seeded (fixed seed = 42) for reproducibility, per CLAUDE.md's pinned-dependency rule.]

3.3 Evaluation Protocol

Each (dataset, model) pair is evaluated with 5-fold cross-validation (fixed seed = 42). Folds are grouped by distinct input recipe (rounded to 6 decimal places) so that duplicate rows sharing the same colorant input always fall in the same fold, preventing train/test leakage through repeated measurements. Within each fold:

1. A `MinMax` scaler is fit on the training fold's inputs and, separately, on the training fold's XYZ targets – scaling parameters are never estimated from the held-out fold.
2. The model is fit in the normalized (0–1) input/output space.
3. Predictions are generated in normalized space, then inverse-transformed back to physical XYZ using the training fold's target scaler.
4. Predicted XYZ values are clipped at zero (tristimulus values cannot be negative).
5. Both predicted and ground-truth XYZ are converted to CIE Lab under the D50 illuminant and 2° standard observer, and the per-sample color difference is computed with the CIE 2000 formula [2] via the `colour-science` library.

Every sample in a dataset is held out exactly once across the 5 folds, so the reported statistics pool over the entire dataset rather than a single train/test split. Per repository convention (see CLAUDE.md), summary statistics are the **median**, **95th percentile**, and **maximum** of the per-sample ΔE_{00} distribution – not the mean or standard deviation, since color-difference errors are not normally distributed and a small number of large outliers otherwise dominate a mean-based summary.

As a data-integrity check, every dataset's own measured XYZ is round-tripped through the same D50/2°/CIE-2000 conversion and compared against its own measured Lab before any model is fit; a median absolute Lab discrepancy above 0.6 aborts the run. This tripwire is a direct response to the scale error described below, and guards against future recurrences (e.g. accidentally feeding normalized XYZ into the Lab conversion).

Correction to the AIC 2025 conference results. The AIC 2025 conference version of this analysis computed ΔE_{00} on normalized (0–1 scaled) XYZ values rather than on the physical 0–100 XYZ scale, which understated the reported color differences by approximately a factor of five. All results reported in this paper are corrected and supersede the conference figures.

4 Results

4.1 $n = 3$ (CMY) – Corrected

Table 3 reports median, 95th-percentile, and maximum ΔE_{00} for all fourteen models on the three $n = 3$ (CMY) datasets, pooled across the 5 cross-validation folds so that every one of the 818 $K = 0$ samples per dataset is predicted exactly once. Figure 1 shows the same medians as a ranked bar chart across the three datasets on a log scale, with a dashed reference line at $\Delta E_{00} = 1$ marking the conventional just-noticeable-difference threshold. Gaussian process regression is the best-median model on all three datasets by a wide margin, followed by third-order polynomial regression as a clear second; the ridge, PCR, PLSR, lasso, and elastic-net family cluster together with median ΔE_{00} around 6.3–6.7, an order of magnitude worse than GP and roughly 20× the just-noticeable-difference threshold.

Table 3: Median, 95th-percentile, and maximum ΔE_{00} per model for the $n = 3$ (CMY) condition on the three printer datasets, 818 $K = 0$ samples each. All values from 5-fold cross-validation with every sample predicted exactly once; ΔE_{00} computed on denormalized XYZ (D50 illuminant, 2° standard observer), per Section 3.3. Models are sorted by PC10 median. Source: `journal/results/{PC10,PC11,FOGRA51}-CMY/summary.csv`.

	PC10			PC11			FOGRA51		
Model	Median	P95	Max	Median	P95	Max	Median	P95	Max
Gaussian Process	0.054	0.189	2.061	0.054	0.160	1.797	0.065	0.191	0.807
Polynomial (3rd order)	0.279	0.966	7.695	0.244	0.890	7.107	0.368	1.163	8.008
SVM	0.754	1.947	5.442	0.717	1.836	4.889	0.766	1.838	6.168
Gradient Boosting	0.881	3.133	22.659	0.791	2.821	13.192	0.819	2.926	17.404
MLP (Deep)	1.102	4.055	15.958	1.054	3.736	25.396	1.176	4.051	23.968
MLP (Shallow)	1.195	4.483	21.150	1.176	4.001	13.442	1.408	6.086	18.983
Random Forest	1.716	4.187	9.264	1.694	4.108	8.467	1.804	4.486	8.131
k-NN	2.012	4.270	8.918	1.938	4.085	8.852	1.867	4.345	9.512
Decision Tree	4.369	9.828	25.211	4.136	8.891	24.132	4.320	9.001	20.566
Lasso	6.601	24.791	43.961	6.292	23.462	45.338	6.658	25.221	42.062
Elastic Net	6.611	24.781	43.890	6.284	23.952	45.261	6.692	25.279	41.981
Ridge	6.624	24.802	43.820	6.309	23.943	45.184	6.719	25.397	41.897
PCR	6.624	24.802	43.820	6.309	23.943	45.184	6.719	25.397	41.897
PLSR	6.643	24.901	43.823	6.307	23.989	45.184	6.715	25.528	41.892

4.2 $n = 4$ (CMYK) and the n=3 → n=4 Comparison

Table 4 reports the same statistics for the $n = 4$ (CMYK) condition, using the full 1,617-sample set (the 818 $K = 0$ rows of Table 3 plus the 799 $K > 0$ rows, with K retained as a fourth input rather than dropped – see the Methods correction paragraph). Gaussian process regression remains the best-median model by a wide margin on all three datasets, and third-order

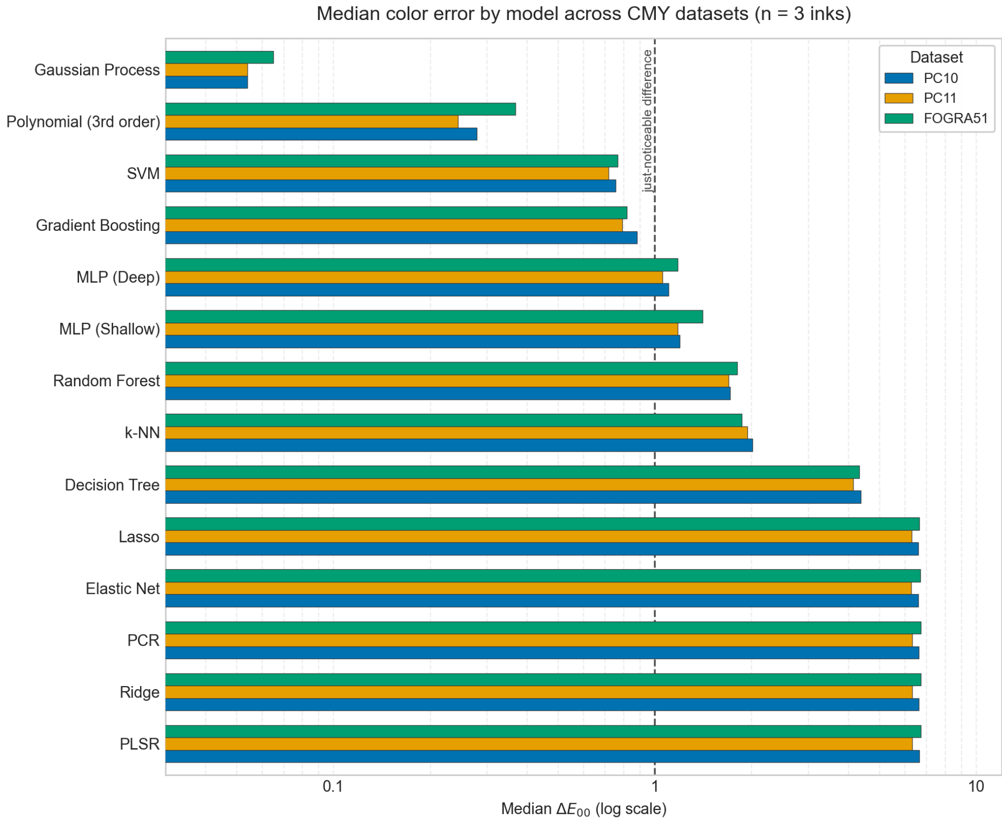

Figure 1: Median ΔE_{00} per model across the three CMY datasets (818 $K = 0$ samples each, 5-fold cross-validation), ranked by mean median across datasets. Log scale; the dashed line marks the just-noticeable difference ($\Delta E_{00} \approx 1$).

polynomial regression remains second; the relative ordering of the remaining models is otherwise stable between $n = 3$ and $n = 4$ (SVM third, gradient boosting fourth, on all three datasets), but every model's absolute error grows, and by very different factors – the subject of Figure 2.

Figure 2 compares median ΔE_{00} at $n = 3$ (CMY) versus $n = 4$ (CMYK) for every model on PC10, the full 1,617-sample set now including the $K > 0$ rows. Adding the fourth input channel is not uniformly costly: third-order polynomial regression, which has no mechanism to represent the extra channel's interaction terms beyond its fixed cubic expansion, degrades from a median of 0.279 to 0.944 ($\times 3.4$), whereas Gaussian process regression is comparatively robust, degrading only from 0.054 to 0.063 ($\times 1.2$). This asymmetry is consistent with GP regression's non-parametric kernel adapting to the additional input dimension, while the fixed-order polynomial basis cannot.

4.3 Gaussian Process vs. Noise Floor

Gaussian process regression is the best-median model in every dataset/condition combination reported above. Because GP is also the model most exposed to a subtle cross-validation leak – duplicate colorant-input rows with distinct measured XYZ landing in both the train and test folds – its result was re-verified with a duplicate-grouped cross-validation split (folds assigned by distinct input recipe rather than by row, so all duplicates of a given recipe fall in the same fold; see Section 3.3). Across all six dataset/condition combinations, GP's median ΔE_{00} ranges from 0.052 to 0.073, and the grouped-CV median never differs from the plain k-fold median by more than a factor of 1.02, confirming the result is not an artifact of duplicate leakage. Third-order polynomial regression, re-checked the same way, is similarly stable. Source:

Table 4: Median, 95th-percentile, and maximum ΔE_{00} per model for the $n = 4$ (CMYK) condition on the three printer datasets, all 1,617 samples (818 $K = 0$ plus 799 $K > 0$) each. Same protocol as Table 3. Models are sorted by PC10 median. Source: `journal/results/{PC10,PC11,FOGRA51}-CMYK/summary.csv`.

	PC10			PC11			FOGRA51		
Model	Median	P95	Max	Median	P95	Max	Median	P95	Max
Gaussian Process	0.063	0.473	8.370	0.066	0.435	6.796	0.072	0.292	3.248
Polynomial (3rd order)	0.944	8.154	32.059	0.868	7.724	32.223	0.822	6.643	28.377
SVM	1.029	5.585	19.663	0.993	4.856	15.270	1.006	4.419	11.337
Gradient Boosting	1.528	6.515	26.210	1.389	6.289	22.283	1.539	6.910	30.171
MLP (Deep)	1.766	11.309	34.578	1.612	9.743	39.302	1.639	10.603	28.127
MLP (Shallow)	1.986	11.103	29.006	1.847	9.657	25.770	1.955	9.976	28.828
k-NN	2.233	5.806	13.515	2.135	5.523	12.148	2.151	5.917	20.065
Random Forest	2.274	4.952	9.182	2.244	4.772	10.348	2.567	5.808	16.117
Decision Tree	4.605	9.679	24.345	4.352	9.184	23.095	4.464	10.227	19.473
Lasso	8.908	24.367	42.965	8.467	23.842	42.354	8.771	25.522	43.085
PCR	8.929	24.526	42.666	8.471	23.459	42.040	8.820	25.102	42.406
Ridge	8.929	24.526	42.666	8.471	23.459	42.040	8.820	25.102	42.406
Elastic Net	8.937	24.480	42.828	8.500	23.651	42.212	8.790	25.174	42.785
PLSR	8.940	25.216	42.695	8.630	23.766	42.028	8.699	25.352	43.411

`journal/results/gp_verification.csv`.

4.4 IFRA Newsprint: Within-Run, Cross-Run, and Leave-One-Out

Three experiments probe how well a model characterizing one printer generalizes to another, using the 13 IFRA white-backing (wb) newsprint press runs described in Section 3.1 (1,485 CMYK samples per run). A model subset (Gaussian process, third-order polynomial, SVM, deep MLP – winner, classical baseline, and two mid-field representatives from Table 4) is evaluated under three training conditions: (A) *within-run*, 5-fold cross-validation on a single press run, the best-case accuracy attainable when a model is fit to that specific press; (B) *cross-run*, trained on one run and tested on a different run with no shared training data (all $13 \times 12 = 156$ ordered press pairs); and (C) *leave-one-run-out*, trained on the pooled data from 12 runs and tested on the 13th, held-out run.

Table 5: Median ΔE_{00} per model across the three IFRA generalization experiments (13 white-backing press runs). Within-run and leave-one-out values are the median of each run's own median ΔE_{00} across the 13 runs; cross-run values are the median across the 156 ordered train/test press pairs. Source: `journal/results/ifra/{within_run, cross_run,leave_one_out}.csv`.

Model	Within-run (A)	Cross-run (B)	Leave-one-out (C)
SVM	1.07	3.87	3.01
Polynomial (3rd order)	1.42	4.17	3.30
MLP (Deep)	1.61	4.13	3.63
Gaussian Process	–[1]	3.89	3.03

[1] The Gaussian process regression's within-run result (median $\Delta E_{00} \approx 18.8$, more than an order of magnitude worse than every other model) is anomalous – a fixed `WhiteKernel` noise-floor setting tuned for the coated-paper datasets is misconfigured for newsprint's substantially higher measurement noise – and is therefore omitted from the within-run comparison.

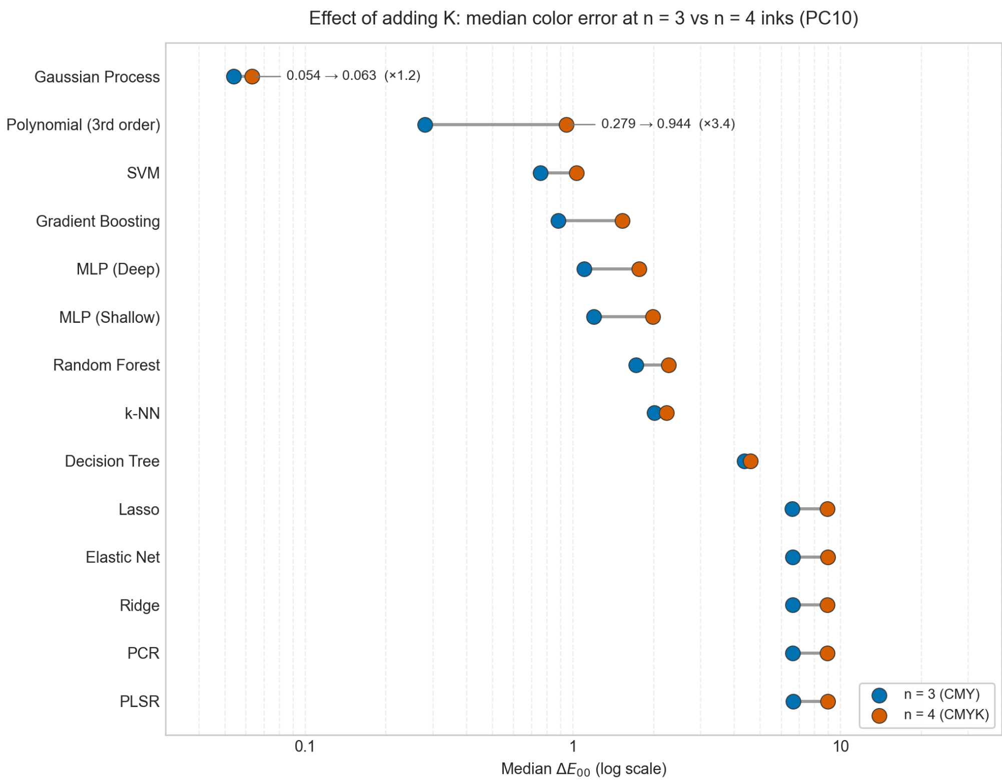

Figure 2: Median ΔE_{00} per model on PC10 at $n = 3$ (CMY, 818 samples) versus $n = 4$ (CMYK, 1,617 samples). Polynomial regression degrades $\times 3.4$ with the added K channel, while Gaussian process regression degrades only $\times 1.2$, the two highlighted comparisons.

The story is that press-to-press variation dominates model choice: every model's cross-run error (B, $\approx$3.9 median regardless of model) is roughly 3–4$\times$ its own within-run error (A), and no amount of model sophistication closes that gap – Gaussian process regression, the clear within-run winner on every coated-paper dataset in Tables 3–4, is statistically indistinguishable from SVM, polynomial regression, and a deep MLP once training and test data come from different presses. Pooling training data across the other 12 runs (C) recovers some of that loss – leave-one-out error ($\approx$3.0–3.6 median) is consistently lower than single-run cross-run transfer (B) for every model – but pooling multiple presses still cannot reach single-press within-run accuracy (A): more data from the wrong press substitutes for, but does not equal, data from the right press. Figure 3 shows the three conditions side by side.

4.5 Direct ΔE_{00} Loss

Every polynomial fit reported so far minimizes ordinary least squares on normalized XYZ, a proxy for the perceptual ΔE_{00} metric the paper actually evaluates on. This experiment tests whether refining the third-order polynomial's coefficients directly on mean ΔE_{00} closes any of that gap. Starting from the least-squares solution, the 60 (CMY) or 105 (CMYK) coefficients are refined with a derivative-free optimizer whose objective is mean ΔE_{00} on the training fold, evaluated in real, denormalized XYZ; the refined model is evaluated with the same grouped 5-fold cross-validation as every other model. Two optimizers are compared: Nelder-Mead (`poly3_de00_nm`, budget 2,000 iterations) and Powell (`poly3_de00_powell`, budget 200 iterations).

Refining directly on ΔE_{00} with Powell reduces worst-case (max) error by 19–48% relative to the least-squares baseline across the three datasets (PC10: -28%, PC11: -19%, FOGRA51: -48%), while the median is essentially unchanged ($\leq$0.011 absolute difference in either direction).

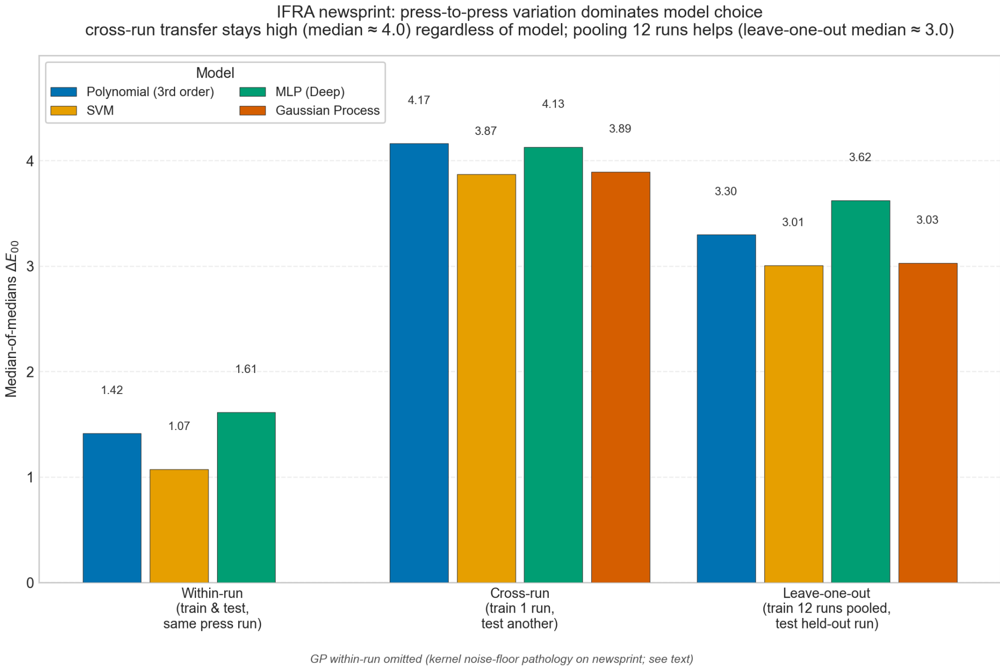

Figure 3: Median ΔE_{00} under the three IFRA generalization conditions (within-run, cross-run, leave-one-run-out) for SVM, third-order polynomial regression, deep MLP, and Gaussian process regression (cross-run and leave-one-out only; see the within-run anomaly note in the text).

Table 6: Median, 95th-percentile, and maximum ΔE_{00} for the least-squares polynomial (`poly3`) versus its two ΔE_{00}-refined variants, $n = 3$ (CMY) condition, three datasets. Source: `journal/results/{PC10,PC11,FOGRA51}-CMY/summary.csv`.

	PC10			PC11			FOGRA51		
Model	Median	P95	Max	Median	P95	Max	Median	P95	Max
poly3 (least squares)	0.279	0.966	7.695	0.244	0.890	7.107	0.368	1.163	8.008
poly3_de00_nm (Nelder-Mead)	0.264	0.982	6.939	0.235	0.847	6.162	0.372	1.170	5.783
poly3_de00_powell (Powell)	0.274	0.886	5.513	0.255	0.772	5.735	0.375	1.043	4.135

Nelder-Mead also reduces max error on all three datasets, but its P95 is inconsistent – worse than the least-squares baseline on PC10 and FOGRA51, better only on PC11 – whereas Powell improves P95 on all three datasets. Powell is therefore the preferred ΔE_{00}-refined variant: it delivers the more consistent, and on FOGRA51 the larger, worst-case improvement, at the cost of a training budget an order of magnitude smaller than Nelder-Mead's (200 vs. 2,000 iterations). The gain is concentrated in the tail of the error distribution rather than the typical case, consistent with ΔE_{00}'s perceptual weighting behaving differently from the uniform weighting of least-squares on XYZ only where errors are already large. Figure 4 shows the three variants' error distributions side by side.

4.6 LLM-as-Color-Predictor

A separate track tests whether a general-purpose large language model, given color-patch examples entirely in its input context (no fine-tuning, no access to the pipeline's models or code), can act directly as a color predictor. Two OpenAI models, GPT-4o and GPT-4o-mini, were each given 400 CMY→XYZ example patches from PC10-CMY in-context and asked to predict XYZ for 100 held-out query patches, with deterministic (temperature-0) decoding; every raw response was retained and re-scored with the same ΔE_{00} machinery as the classical models. All 100 re-

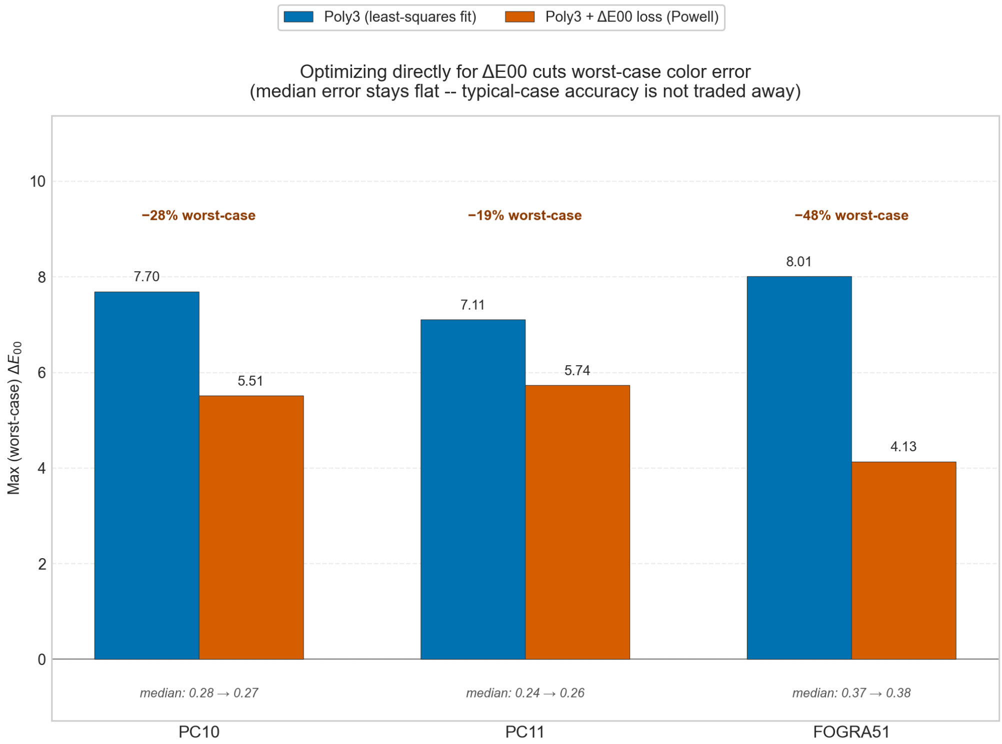

Figure 4: Median/P95/max ΔE_{00} for the least-squares polynomial versus its Nelder-Mead- and Powell-refined variants, across the three $n = 3$ (CMY) datasets.

sponses per model parsed successfully, so the plain and worst-case-penalized scoring variants are identical here.

Table 7: Median, 95th-percentile, and maximum ΔE_{00} for two LLMs as direct color predictors, PC10-CMY, 100 held-out query patches with 400 in-context training examples. Source: `journal/results/llm/PC10-CMY_summary.csv`.

Model	Median	P95	Max
GPT-4o	3.034	10.642	14.621
GPT-4o-mini	9.445	29.928	44.792

A frontier LLM predicts XYZ from in-context examples with visible error (ΔE_{00} median 3.034 for GPT-4o, roughly three times the just-noticeable-difference threshold): it lands mid-field against the classical/ML model set of Table 3, beating the decision tree (median 4.369) and the entire ridge/lasso/elastic-net/PCR/PLSR linear family (medians 6.6–6.7) but losing to every other model, including the k-nearest-neighbors baseline (median 2.012). It is $\approx$11$\times$ worse than third-order polynomial regression (median 0.279) and $\approx$56$\times$ worse than Gaussian process regression (median 0.054), the paper's best model. The smaller GPT-4o-mini is uniformly worse than every classical method (median 9.445, worse even than the linear family).

This comparison is not apples-to-apples with the rest of the paper, and that methodological gap should be stated explicitly: the classical/ML results use 5-fold cross-validation over 818 rows, so every sample is held out and scored exactly once, whereas the LLM results are a single 100-sample holdout scored against 400 in-context examples, with no cross-validation – the LLM numbers therefore carry more sampling uncertainty and are not directly comparable in rigor to the classical column of Table 3, even though both are computed with the identical ΔE_{00} formula.

Figure 5 places the LLM results alongside the classical/ML models for visual comparison, with this caveat noted in the caption.

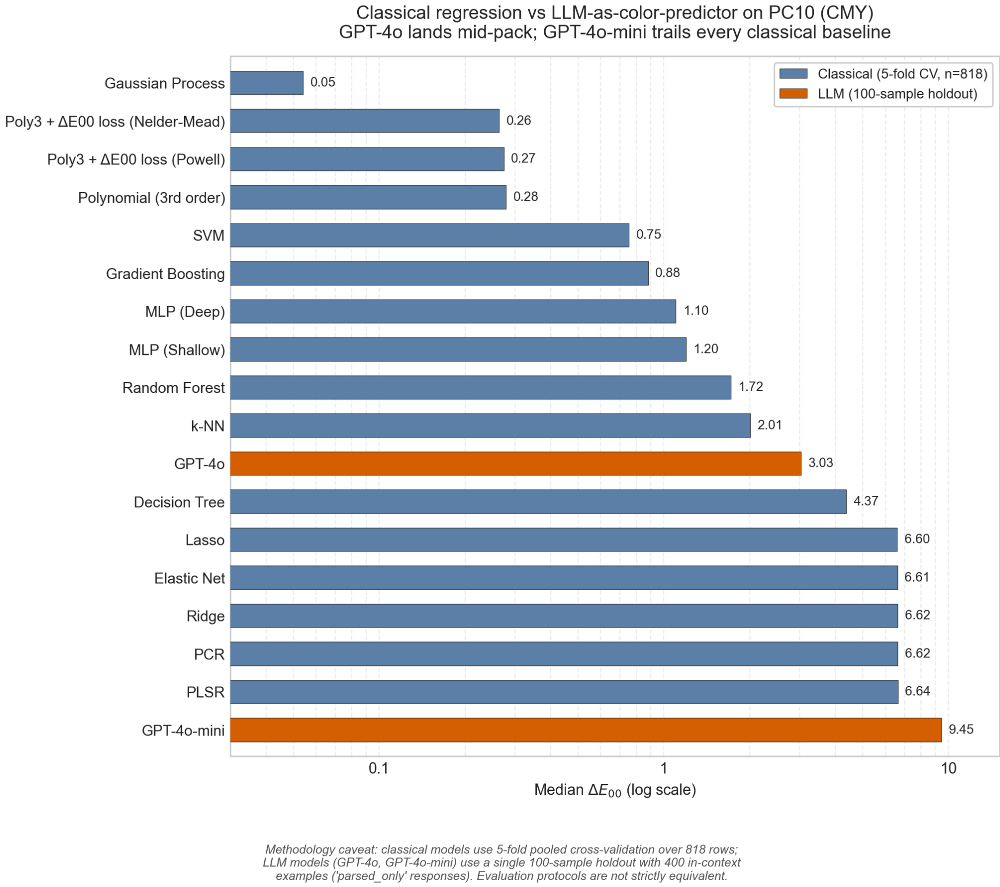

Figure 5: Median ΔE_{00} for GPT-4o and GPT-4o-mini as direct color predictors (single 100-sample holdout, 400 in-context examples) alongside the 14 classical/ML models (5-fold cross-validation, 818 samples), PC10-CMY. The two evaluation protocols differ as described in the text; log scale, dashed line at $\Delta E_{00} = 1$.

4.7 $n = 7$ (CMYKOGV)

[RESULTS-PENDING: Plan 06 results, conditional on the dataset arriving from Phil in time. Figure F6 (median ΔE_{00} vs. $n \in \{3, 4, 7\}$).]

5 Discussion

[TODO: Task 2. Answer Phil's two core questions explicitly: (a) does any ML method beat classical polynomial regression at $n \leq 4$, and if so by how much, avoiding gray-component-replacement-style handling of K; (b) does any method handle $n > 4$ successfully where polynomial regression is not designed to scale (contingent on the $n = 7$ dataset). Limitations to cover: single-substrate training sets (no cross-printer generalization claim without the IFRA results); GP's $O(N^3)$ training-set-size scaling as a practical deployment concern; the K-ambiguity lesson from the AIC 2025 conference version as a cautionary methodological note (kept minimal, per author decision – see Methods correction paragraph – no separate pitfalls section unless Phil asks for one on review).]

6 Conclusion

[TODO: Task 2. One paragraph, restating the corrected headline findings and, if landed, the $n > 4$ finding.]

Data Availability Statement

Code and data supporting this study are available at https://github.com/hamzafer/Evaluating-Machine-Lea
[TODO: Confirm this is still the correct public repository URL before submission (ported from the AIC 2025 conference paper's Code and Data Availability section) and that it reflects the corrected pipeline described here, not only the legacy AIC v1 code.]

References

[1] Fogra Research Institute. FOGRA51 and FOGRA52 characterisation data (version 1.0) [data set]. https://fogra.org/en/downloads/work-tools/characterisation-data, 2020.

[2] Gaurav Sharma, Wencheng Wu, and Edul N. Dalal. The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research & Application*, 30(1):21–30, 2005.